

Mathemalchemy Temari Ball Arches and Infinite Series

Consider two infinite sequences of spheres placed along a straight line, with each sphere tangent to the next. Let the diameters of the spheres be given by the two sequences

$$D_n = D_1 \left(\frac{4}{n+3} \right)^{2/3}, \quad n = 1, 2, 3, \dots,$$

and

$$d_n = d_1 \left(\frac{4}{5} \right)^{\frac{2(n-1)}{3}}, \quad n = 1, 2, 3, \dots$$

Suppose that

$$D_1 = d_1.$$

Thus the two constructions begin with exactly the same sphere. They also have the same second sphere, since

$$D_2 = D_1 \left(\frac{4}{5} \right)^{2/3}$$

and

$$d_2 = d_1 \left(\frac{4}{5} \right)^{2/3}.$$

Therefore,

$$D_1 = d_1 \quad \text{and} \quad D_2 = d_2.$$

After the second sphere, however, the two sequences shrink according to different rules, and their long-term behavior is very different.

1. The convergent diameter series

For the d_n sequence,

$$d_n = d_1 \left(\frac{4}{5} \right)^{\frac{2(n-1)}{3}}.$$

This is a geometric sequence with common ratio

$$r = \left(\frac{4}{5} \right)^{2/3}.$$

Since $0 < r < 1$, the series of diameters converges:

$$\sum_{n=1}^{\infty} d_n = d_1 \sum_{n=1}^{\infty} \left(\frac{4}{5} \right)^{\frac{2(n-1)}{3}}.$$

Therefore,

$$\boxed{\sum_{n=1}^{\infty} d_n = \frac{d_1}{1 - \left(\frac{4}{5}\right)^{2/3}}}$$

is finite.

If these spheres are laid side by side, tangent to one another along a straight line, then infinitely many spheres fit within a finite total length.

Numerically,

$$\frac{1}{1 - \left(\frac{4}{5}\right)^{2/3}} \approx 7.2345,$$

so the total length of the chain is approximately

$$7.2345 d_1.$$

2. The divergent diameter series

Now consider

$$D_n = D_1 \left(\frac{4}{n+3} \right)^{2/3}.$$

The corresponding series of diameters is

$$\sum_{n=1}^{\infty} D_n = 4^{2/3} D_1 \sum_{n=1}^{\infty} \frac{1}{(n+3)^{2/3}}.$$

Apart from a constant factor and a shift of the index, this is the p -series

$$\sum \frac{1}{n^{2/3}}.$$

Since

$$p = \frac{2}{3} < 1,$$

the series diverges. Therefore,

$$\boxed{\sum_{n=1}^{\infty} D_n = \infty.}$$

Although the individual diameters satisfy

$$D_n \longrightarrow 0,$$

they do not decrease rapidly enough for their sum to converge.

Thus, if these spheres are placed tangent to one another along a straight line, the chain has no finite endpoint. Its total length increases without bound.

3. Volume of a sphere

A sphere with diameter D_n has radius

$$r_n = \frac{D_n}{2}.$$

Its volume is therefore

$$V_n = \frac{4}{3}\pi \left(\frac{D_n}{2}\right)^3 = \frac{\pi}{6}D_n^3.$$

For the divergent diameter sequence,

$$D_n = D_1 \left(\frac{4}{n+3}\right)^{2/3}.$$

Cubing this expression gives

$$D_n^3 = D_1^3 \left(\frac{4}{n+3}\right)^2.$$

Therefore the volume of the n -th sphere is

$$V_n = \frac{\pi}{6}D_1^3 \left(\frac{4}{n+3}\right)^2$$

or

$$V_n = \frac{8\pi D_1^3}{3(n+3)^2}.$$

4. Total volume of the infinitely long chain

The total volume of all the spheres is

$$V_{\text{total}} = \sum_{n=1}^{\infty} V_n.$$

Thus,

$$V_{\text{total}} = \frac{\pi D_1^3}{6} \sum_{n=1}^{\infty} \left(\frac{4}{n+3}\right)^2.$$

Equivalently,

$$V_{\text{total}} = \frac{16\pi D_1^3}{6} \sum_{n=1}^{\infty} \frac{1}{(n+3)^2}.$$

Now the relevant p -series has exponent

$$p = 2 > 1,$$

so it converges. Hence

$$\sum_{n=1}^{\infty} V_n < \infty.$$

This is the striking feature of the construction:

The chain has infinite total length but finite total volume.

The diameters decrease too slowly for their sum to converge, but volume depends on the *cube* of the diameter. In particular,

$$D_n \sim \frac{1}{n^{2/3}},$$

while

$$V_n \sim D_n^3 \sim \left(\frac{1}{n^{2/3}}\right)^3 = \frac{1}{n^2}.$$

Thus

$$\sum D_n \sim \sum \frac{1}{n^{2/3}}$$

diverges, whereas

$$\sum V_n \sim \sum \frac{1}{n^2}$$

converges.

5. The exact limiting volume

Because

$$\sum_{n=1}^{\infty} \frac{1}{(n+3)^2} = \sum_{k=4}^{\infty} \frac{1}{k^2},$$

we can use

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

Therefore,

$$\sum_{k=4}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6} - \left(1 + \frac{1}{4} + \frac{1}{9}\right).$$

Since

$$1 + \frac{1}{4} + \frac{1}{9} = \frac{49}{36},$$

we obtain

$$V_{\text{total}} = \frac{8\pi D_1^3}{3} \left(\frac{\pi^2}{6} - \frac{49}{36}\right).$$

If V_1 denotes the volume of the first sphere, then

$$V_1 = \frac{\pi}{6} D_1^3.$$

Consequently,

$$\frac{V_{\text{total}}}{V_1} = 16 \left(\frac{\pi^2}{6} - \frac{49}{36}\right) \approx 4.54117.$$

Thus the total volume of the entire infinitely long chain is only about

$$V_{\text{total}} \approx 4.54117 V_1.$$

In other words, an infinite chain of spheres stretching arbitrarily far along a line contains only about 4.54 times the volume of its first sphere.

6. Comparison with the convergent chain

The d_n construction is even more strongly convergent. Since

$$d_n = d_1 \left(\frac{4}{5} \right)^{\frac{2(n-1)}{3}},$$

cubing gives

$$d_n^3 = d_1^3 \left(\frac{4}{5} \right)^{2(n-1)} = d_1^3 \left(\frac{16}{25} \right)^{n-1}.$$

The volumes therefore form a geometric series:

$$\sum_{n=1}^{\infty} \frac{\pi}{6} d_n^3 = \frac{\pi d_1^3}{6} \sum_{n=1}^{\infty} \left(\frac{16}{25} \right)^{n-1}.$$

Hence

$$V_{\text{total},d} = \frac{\pi d_1^3}{6} \frac{1}{1 - \frac{16}{25}} = \frac{25}{9} \left(\frac{\pi d_1^3}{6} \right).$$

If the first sphere has volume V_1 , then

$$V_{\text{total},d} = \frac{25}{9} V_1 \approx 2.7778 V_1.$$

Thus the two constructions provide an interesting comparison:

Sphere diameters	Total length	Total volume
d_n	finite	finite
D_n	infinite	finite

The second case illustrates an important feature of infinite series: different geometric quantities associated with the same objects can have completely different convergence behavior.